Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the f1 score during training.

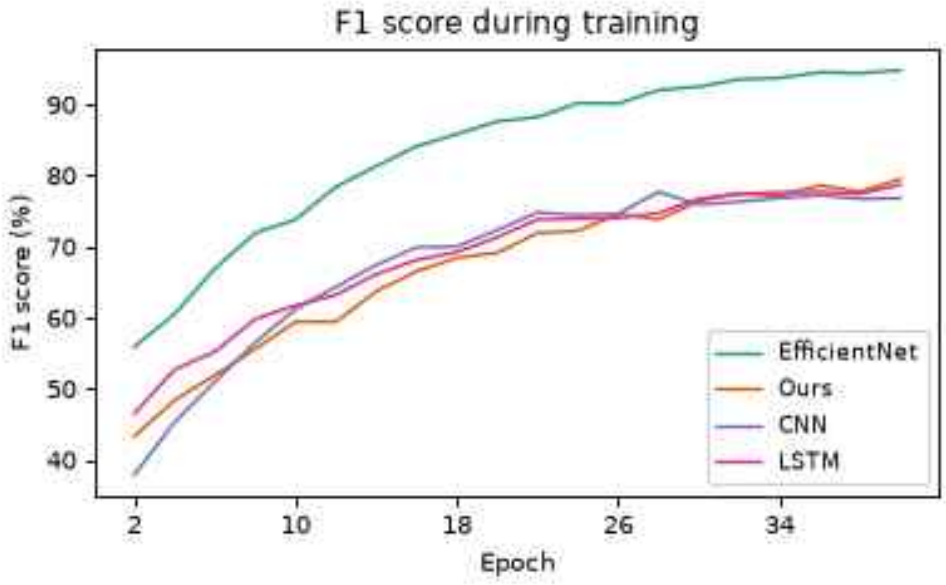


Figure 1: F1 score during training for the models in this experiment.

As shown in Figure 1, at epoch 40, EfficientNet is 15.3 points better than Ours. As shown in Figure 1, from epoch 16 to epoch 26, the F1 score of CNN improves by a relative 15.7%. At epoch 6, EfficientNet is 3.4 points better than CNN (Figure 1). As shown in Figure 1, from epoch 2 to epoch 4, the F1 score of LSTM improves by a relative 10.6%.

Figure 2 shows the validation accuracy of BERT-base by dataset.

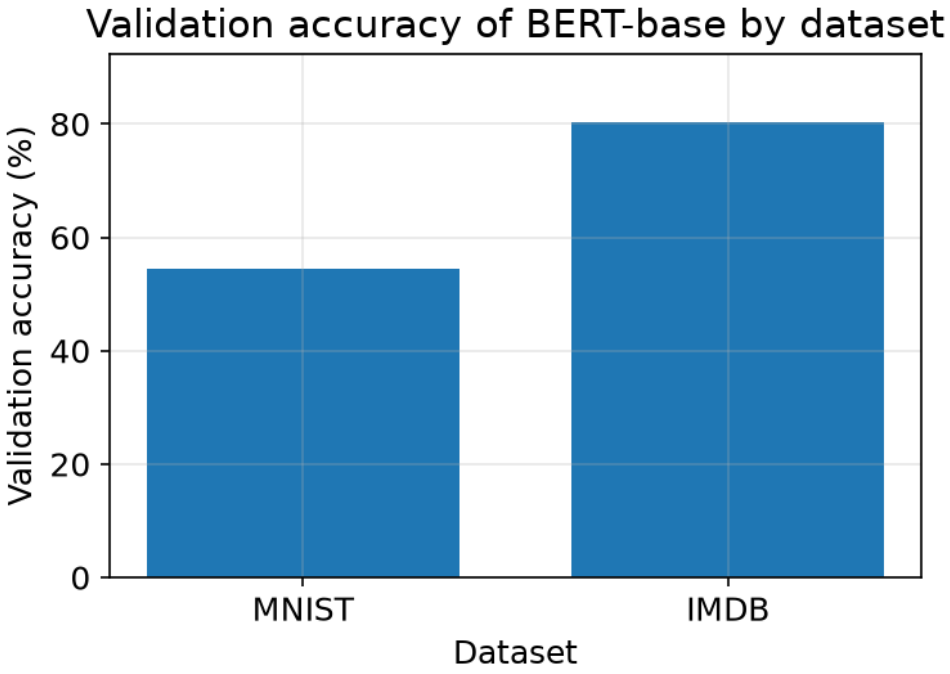


Figure 2: Validation accuracy of BERT-base by dataset for the models in this experiment.

BERT-base reaches over 50% validation accuracy on MNIST. BERT-base achieves 63.7% validation accuracy on MNIST. As shown in Figure 2, BERT-base reaches 45.3% validation accuracy on MNIST.

Figure 3 shows the validation accuracy during training.

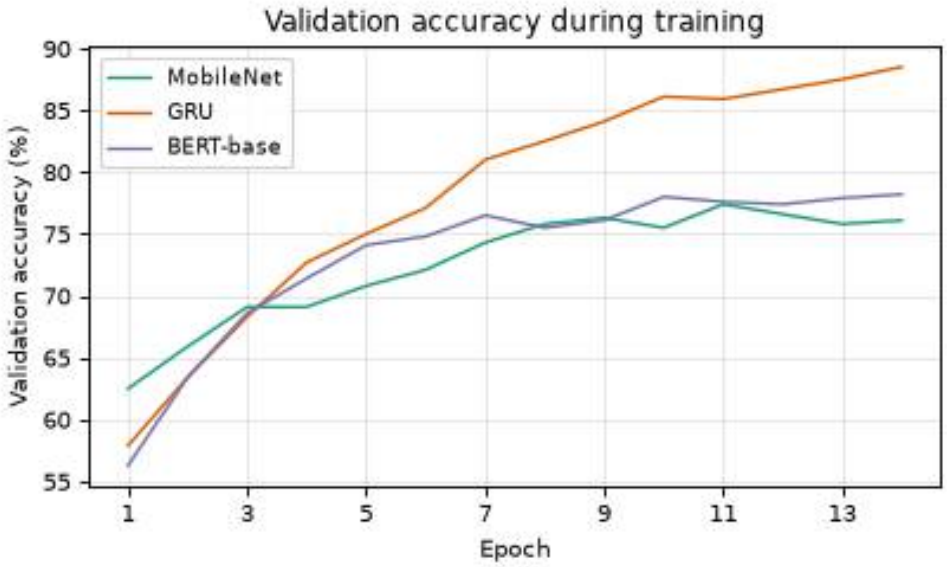


Figure 3: Validation accuracy during training for the models in this experiment.

As shown in Figure 3, from epoch 7 to epoch 8, the validation accuracy of GRU improves by a relative 2%. GRU attains a validation accuracy of 75.8% at epoch 13 (Figure 3).

3 Discussion

On QQP, Random Forest obtains an accuracy of 63.8%. The differences between the models are discussed in the next section.